WORKED SOLUTION

STEP II 2014, Question 11

Mechanics • Circular Motion

STEP II 2014, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q11

The key idea

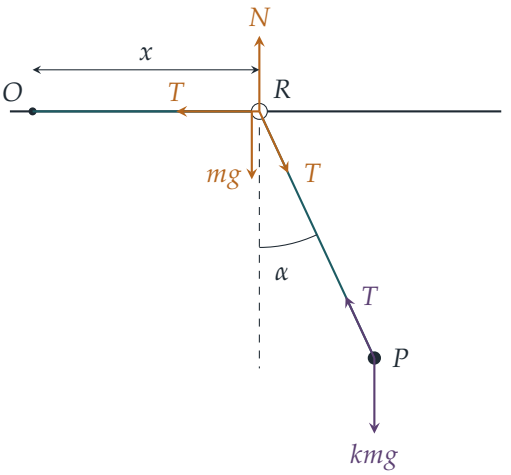
Everything is driven by one variable, x. Because the string is inextensible and α is fixed, the position of P is a simple function of x, so both of P's acceleration components are multiples of $\ddot{x}$. That gives three equations of motion (two for P, one horizontal one for the ring) in just two unknowns, $\ddot{x}$ and T. An overdetermined system is only consistent if something special holds, and that consistency condition is what ties α to k.

Solution

Model. The ring and P are particles, the string is light and inextensible, and the ring and rail are smooth. Since the ring is smooth, the tension T is the same in both parts of the string. Take O as origin, the x-axis along the rail in the direction from O to the ring, and y vertically up.

Coordinates of P. The part of the string from O to the ring lies along the rail and has length x, so the part from the ring to P has length $L - x$. It makes angle α with the downward vertical, and P hangs on the side of the ring away from O (we justify this choice at the end of part (i)). So

$$P = (x + (L - x)\sin\alpha,\ -(L - x)\cos\alpha).$$



Part (i)

Differentiating the coordinates of P twice, with α constant, gives

$$\ddot{x}_P = \ddot{x}\,(1 - \sin\alpha), \qquad \ddot{y}_P = \ddot{x}\cos\alpha.$$

The forces on P are its weight kmg downwards and the tension T along the string towards the ring, whose components are $-T\sin\alpha$ horizontally and $+T\cos\alpha$ vertically.

P, vertically: $km\,\ddot{y}_P = T\cos\alpha - kmg$, that is

$$km\ddot{x}\cos\alpha = T\cos\alpha - kmg,$$

as required.

P, horizontally:

$$km\ddot{x}\,(1 - \sin\alpha) = -T\sin\alpha. \tag{1}$$

Ring, horizontally. The ring feels the string on both sides: the part along the rail pulls it towards O with force T, and the part going down to P pulls it with horizontal component $T\sin\alpha$ away from O. Its weight and the normal reaction of the rail are vertical (the rail is smooth). So

$$m\ddot{x} = T\sin\alpha - T = -T(1 - \sin\alpha). \tag{2}$$

? **Ask yourself** Why must P be on the far side of the ring from O?

If P hung on the O side, its x-coordinate would be $x - (L - x)\sin\alpha$, and the two horizontal equations would become $km\ddot{x}(1 + \sin\alpha) = T\sin\alpha$ and $m\ddot{x} = -T(1 + \sin\alpha)$. The first makes $\ddot{x} \geq 0$ and the second makes $\ddot{x} \leq 0$, so $\ddot{x} = 0$ and $T = 0$; but then the vertical equation gives $0 = -kmg$, which is impossible. So that configuration can't move with constant α, and the coordinates above are the right ones.

Part (ii)

First, $T \neq 0$: if $T = 0$ then (2) gives $\ddot{x} = 0$, and the vertical equation gives $0 = -kmg$, a contradiction.

From (2), $m\ddot{x} = -T(1 - \sin\alpha)$. Substituting this into (1):

$$k(-T(1 - \sin\alpha))(1 - \sin\alpha) = -T\sin\alpha.$$

Dividing by $-T \neq 0$,

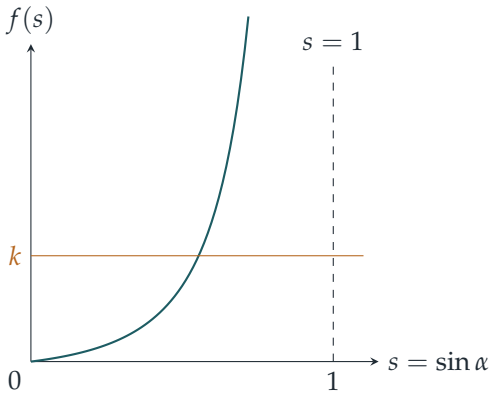
$$\frac{\sin\alpha}{(1 - \sin\alpha)^2} = k,$$

as required. (Since α is acute, $0 < \sin\alpha < 1$, so the division by $(1 - \sin\alpha)^2$ is fine.)

Every k is possible. Put $s = \sin\alpha$ and $f(s) = \dfrac{s}{(1-s)^2}$ on $0 < s < 1$, which covers exactly the acute angles α with $\alpha \neq 0$. By the quotient rule,

$$f'(s) = \frac{(1-s)^2 + 2s(1-s)}{(1-s)^4} = \frac{1+s}{(1-s)^3} > 0,$$

so f is strictly increasing. Also $f(s) \to 0$ as $s \to 0^+$ and $f(s) \to \infty$ as $s \to 1^-$, and f is continuous. So f takes every positive value exactly once, and for each $k > 0$ there is exactly one acute angle α satisfying the condition.



Whatever the height of the horizontal line $y = k$, it meets the curve exactly once.

Part (iii)

We now eliminate T using (2) and the vertical equation. From (2), $T = -\dfrac{m\ddot{x}}{1 - \sin\alpha}$. Substituting into the vertical equation and dividing by m:

$$k\ddot{x}\cos\alpha = -\frac{\ddot{x}\cos\alpha}{1 - \sin\alpha} - kg \quad \Longrightarrow \quad \ddot{x}\cos\alpha\left(k + \frac{1}{1 - \sin\alpha}\right) = -kg.$$

Now use part (ii) to replace k inside the bracket:

$$k + \frac{1}{1-\sin\alpha} = \frac{\sin\alpha + (1-\sin\alpha)}{(1-\sin\alpha)^2} = \frac{1}{(1-\sin\alpha)^2}.$$

Hence

$$\ddot{x} = -\frac{kg(1-\sin\alpha)^2}{\cos\alpha} = -\frac{g\sin\alpha}{\cos\alpha} = -g\tan\alpha,$$

using $k(1-\sin\alpha)^2 = \sin\alpha$ once more. This is the required result.

As a bonus, $T = -\dfrac{m\ddot{x}}{1-\sin\alpha} = \dfrac{mg\tan\alpha}{1-\sin\alpha} > 0$, so the string really is taut, as the model needs.

Checks

- With $k = 1$, the condition $s = (1-s)^2$ gives $\sin\alpha = \frac{1}{2}(3-\sqrt{5}) \approx 0.382$, so $\alpha \approx 22.46^\circ$ and $\ddot{x} \approx -0.413g$. Substituting into (1), (2) and the vertical equation, all three balance.
- We simulated the full two-variable motion (in x and the string angle) numerically, starting at this angle with the ring moving: the angle stayed constant to within 10^{-10} rad and the ring decelerated at exactly $g\tan\alpha$. Starting at a slightly different angle, the angle immediately begins to change, so the special value of α really matters.
- Limiting cases: as $k \to 0$ (a very light particle), $\alpha \to 0$ and $\ddot{x} \to 0$; as $k \to \infty$, $\alpha \to 90^\circ$ and the deceleration grows without bound, because the light ring is being hauled by a heavy particle.
- Signs: $\ddot{x} < 0$, so the ring is always accelerating towards O. That fits: the net horizontal force on the ring, $-T(1-\sin\alpha)$, points towards O.

Common slips

- Forgetting that the ring feels the string on *both* sides; the pull along the rail towards O is what makes (2) work.
- Writing the horizontal acceleration of P as $\ddot{x}$; the length $L - x$ of the hanging part also changes, which contributes $-\ddot{x}\sin\alpha$.
- Treating α as a variable when differentiating. It is constant here, which is exactly what makes the problem tractable.

Beyond STEP

This is a system with two degrees of freedom (the ring's position and the string's angle), and the question looks for a special solution in which one of them stays fixed, in the spirit of a relative equilibrium. In a first-year course you would write down the Lagrangian in x and the angle and find that constant-angle motion is possible only for

the one value of α found here; the natural next question is whether that motion is stable. Kleppner and Kolenkow, *An Introduction to Mechanics*, is good on constrained systems of this kind.

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